

Part 1 of 2

Low-Power MCU Selection

Power consumption is very important for battery-powered applications where the battery must last for years. This article discusses key low-power features of MCUs and hardware/software techniques to achieve intelligent power management

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With the increasing use of microcontroller units (MCUs) in power-conscious applications, low power has become a very important parameter to consider while selecting them. New low-power 8-/16-/32-bit MCUs designed for portable battery-powered products include an impressive set of features. These are available from many companies, which makes it difficult to choose the best. Proper use of low-power mode (LPM) provided in most MCUs is also very critical.

To select the right MCU, the designer must understand LPMs, clocking systems, event-driven capability, on-chip peripherals, brownout detection (BOD), pin leakage current and processing efficiency.

Low-power MCU parameters

Average current consumption. Power consumption is the product of operating voltage (V_{cc}) multiplied by current consumption (I_{cc}). Usually, only I_{cc} is mentioned while describing power characteristics of a chip. This is a mistake because decreasing V_{cc} directly reduces the current consumption and overall power drain.

In low-power designs, the average current consumption determines battery life. If an application uses a battery rated at 400 mAh, the application must draw less than 45.7 μ A of average current (400 mAh/8760 hours) to provide one year of battery life.

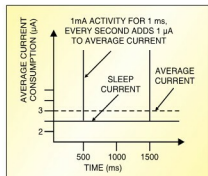


Fig. 1: Average current consumption

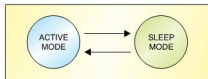


Fig. 2: Active mode versus sleep mode

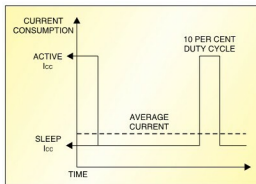


Fig. 3: Power budget

Fig. 1 shows that an application can draw much more current for short time periods keeping the average current consumption below the target.

Low power mode. In many battery-operated applications, the CPU does not run continuously and peripherals

may be idle most of the time. For these applications, sleep mode may consume the lion's share of power and is the vital parameter to consider. The overall power consumption can be lowered keeping the CPU into various LPMs it supports. LPMs provide different levels of functionality to meet the current budget.

In sleep mode, the CPU core, internal memory, DMA controller and external bus interface are switched off, program execution stops, and specific clock supply and associated peripherals continue to run. The biggest power savings are possible only by frequently switching the MCU to an LPM state (Fig. 2) from normal state.

Since an MCU can spend most of the time being inactive, the designer must use a power budget (Fig. 3), i.e., the duty cycle between low-power and active modes, to calculate the average power consumption and battery requirements. Consider an MCU running from a coin cell of 20mAh rating. Typically, the current consumption of an MCU, e.g., MSP430, in its normal power state is 3 mA, whereas it is less than 1 μ A in low-power state. If the MCU is allowed to run in its normal

state at all times, the coin cell will last less than seven hours, but if it is intelligently switched to its lower-power state most of the time, the battery can last more than two years. Thus implementing the power management scheme can dramatically increase the